Algebraic Geometry Problem Set #4

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Problem (Exercise 1)**.** *Let X be a scheme. Let $U, V \subseteq X$ be two affine open subsets. Prove that every $x \in U \cap V$ admits an open neighbourhood $W \subseteq U \cap V$ which is distinguished in both U and V.*

Solution. The intersection $U \cap V$ is an open subset of X contained in U, thus is an open subset of the affine scheme U. In any affine scheme, distinguished opens form a basis of Zariski topology. Therefore, since $x \in U \cap V$, there exists a regular function $f \in \mathscr{O}_X(U)$ such that $x \in D_U(f) \subseteq U \cap V$. Similarly, as $D_U(f)$ is an open subset of the affine scheme V, there exists a regular function $g \in \mathscr{O}_X(V)$ such that $x \in D_V(g) \subseteq D_U(f)$. This way, we got a chain:

$$x \in D_V(g) \subseteq D_U(f) \subseteq U \cap V.$$

We claim $W = D_V(g)$. The hard part is showing that it is distinguished in U.

Since $D_U(f)$ is distinguished open subset of affine scheme U, we know that $\mathscr{O}_X(D_U(f)) \cong \mathscr{O}_X(U)[f^{-1}]$. Also note that we can restrict $g \in \mathscr{O}_X(V)$ to $D_U(f)$: let $\overline{g} := g|_{D_U(f)} \in \mathscr{O}_X(D_U(f))$. Because $\mathscr{O}_X(D_U(f))$ is localized at f, write

$$\overline{g} = \frac{f'}{f^n}, \quad f' \in \mathscr{O}_X(U).$$

Now we do the following non-obvious series of observations:

$$W = D_V(g) = D_{D_U(f)}(\overline{g}) = D_{D_U(f)}(f') = D_U(f) \cap D_U(f') = D_U(ff').$$

First equality is our definition of W. The second is due to the fact that the points where g does not vanish on V is precisely the set of points where $\overline{g}$ does not vanish on $D_U(f)$. The third equality is since for $\overline{g}$ to be zero in $D_U(f)$, the numerator must be zero (as f is inherently invertible in $\mathscr{O}_X(U)[f^{-1}]$). Finally, for the fourth equality notice that points in $D_U(f)$ where f' does not vanish is exactly the set of points where both f and f' do not vanish. The last equality is obvious.

Thus, $W = D_U(ff')$ and since $f, f' \in \mathscr{O}_X(U)$, $ff' \in \mathscr{O}_X(U)$, which proves the statement.

Problem (Exercise 2, Affine communication lemma)**.** *Let X be a scheme and let $\mathscr{U}$ be a family of affine open subsets of X enjoying the following property: for every affine open $U \subseteq X$ and every finite cover $U = U_1 \cup \cdots \cup U_r$ by distinguished opens $U_1, \ldots, U_r \subseteq U$ we have*

$$U \in \mathscr{U} \;\Leftrightarrow\; U_1, \ldots, U_r \in \mathscr{U}.$$

Suppose that X admits an affine open cover $X = \bigcup_{\alpha \in I} U_\alpha$ with $U_\alpha \in \mathscr{U}$. Show that every affine open $U \subseteq X$ belongs to $\mathscr{U}$.

Solution. We first prove the auxilary claim.

Claim. Let $U \in \mathscr{U}$, then any distinguished open $V \subseteq U$ is in $\mathscr{U}$.

Proof. Let $V = D(f) \subseteq U$. Find the distinguished open sets $\{D(g_\alpha)\}_{\alpha \in I}$ that cover $U \setminus V$. Thus, $U = V \cup \bigcup_{\alpha \in I} D(g_\alpha)$. Since U is an affine scheme, it is quasi-compact. Thus, extract a finite subcover: $U = V \cup D(g_1) \cup \cdots \cup D(g_r)$. By the property of $\mathscr{U}$, it immediately follows that $V \in \mathscr{U}$.

Now let $V \subseteq X$ be arbitrary open. We are given the open cover $X = \bigcup_{\alpha \in I} U_\alpha$ and $U_\alpha \in \mathscr{U}$ for all $\alpha \in I$. Pick any point $x \in V$ and assume $x \in U_\beta$ for some β. Thus, $x \in V \cap U_\beta$. Apply Exercise 1: there exists an open W_x such that $x \in W_x \subseteq V \cap U_\beta$ and W_x is distinguished in both V and U_β. By auxilary claim above, as $U_\beta \in \mathscr{U}$, we obtain $W_x \in \mathscr{U}$.

Finally, notice that $V = \bigcup_{x \in V} W_x$ and as V is affine, it is quasi-compact, so $V = W_{x_1} \cup \cdots \cup W_{x_r}$. Since $W_{x_i} \in \mathscr{U}$, we obtain $V \in \mathscr{U}$.

Problem (Exercise 3). *Let X be either a scheme or an algebraic set. Let $f_1, \ldots, f_r \in \mathscr{O}(X)$ be elements such that the opens $D(f_1), \ldots, D(f_r) \subseteq X$ are affine and which generate the unit ideal in $\mathscr{O}(X)$. Show that X is affine.*

Solution. We prove that $\pi : X \to \operatorname{Spec}(\mathscr{O}_X(X))$ is an isomorphism. Given setup, it suffices to prove that restriction $\pi_i : D_X(f_i) \to D_{\mathscr{O}_X(X)}(f_i)$ is an isomorphism of schemes. Finally, since both $D_X(f_i)$ and $D_{\mathscr{O}_X(X)}(f_i)$ are affine, this reduces to showing that induced map of their coordinate rings is an isomorphism: $\mathscr{O}_X(X)[f_i^{-1}] \to \mathscr{O}_X(D(f_i))$.

Since $D(f_1), \ldots, D(f_r)$ cover X, we have an exact sequence

$$0 \to \mathscr{O}_X(X) \to \prod_{j=1}^{r} \mathscr{O}_X(D(f_j)) \to \prod_{j,k=1}^{r} \mathscr{O}_X(D(f_j f_k)).$$

Localization functor is exact, so apply $(-)[f_i^{-1}]$:

$$0 \to \mathscr{O}_X(X)[f_i^{-1}] \to \prod_{j=1}^{r} \mathscr{O}_X(D(f_j))[f_i^{-1}] \to \prod_{j,k=1}^{r} \mathscr{O}_X(D(f_j f_k))[f_i^{-1}].$$

Now notice that $\mathscr{O}_X(D(f_j))[f_i^{-1}] \cong \mathscr{O}_X(D(f_i f_j))$ and similarly $\mathscr{O}_X(D(f_j f_k))[f_i^{-1}] \cong \mathscr{O}_X(D(f_i f_j f_k))$. Thus,

$$0 \to \mathscr{O}_X(X)[f_i^{-1}] \to \prod_{j=1}^{r} \mathscr{O}_X(D(f_i f_j)) \to \prod_{j,k=1}^{r} \mathscr{O}_X(D(f_i f_j f_k))$$

is exact. But note that this is nothing but a sheaf sequence for $\mathscr{O}_{D(f_i)}$ using the open cover $D(f_i f_1), \ldots, D(f_i f_r)$. Thus, $\mathscr{O}_X(X)[f_i^{-1}] \cong \mathscr{O}_X(D(f_i))$ as was required.

Problem (Exercise 4). *Let $f : Y \to X$ be a map of schemes. Prove that f is affine iff exists an affine open cover $X = \bigcup_{\alpha \in I} U_\alpha$ such that $f^{-1}(U_\alpha)$ is affine for each $\alpha \in I$.*

Solution.

($\Rightarrow$) This is trivial. Any scheme admits an affine open cover, say, $\bigcup_\alpha U_\alpha$, and $f^{-1}(U_\alpha)$ is affine by definition of f.

($\Leftarrow$) We employ Exercise 2. Specifically, let

$$\mathscr{U} := \{U \subseteq X : U \text{ is affine open, } f^{-1}(U) \text{ is an affine scheme}\}.$$

We are given the cover $\{U_\alpha\}_{\alpha \in I}$ such that $U_\alpha \in \mathscr{U}$. Assuming $\mathscr{U}$ enjoys the property from Exercise 2, we obtain that every open $U \subseteq X$ is in $\mathscr{U}$, thus $f^{-1}(U)$ is an affine scheme, which is precisely the definition of f being affine.

Now we shall prove why for any affine open $U \subseteq X$ and finite cover $U = U_1 \cup \cdots \cup U_r$ by distinguished opens $U_i = D_U(g_i)$ we have $U \in \mathscr{U} \Leftrightarrow U_1, \ldots, U_r \in \mathscr{U}$.

($\Rightarrow$) Assume $U \in \mathscr{U}$, meaning $f^{-1}(U)$ is an affine scheme. Note:

$$f^{-1}(U_i) = f^{-1}(D_U(g_i)) = D_{f^{-1}(U)}(f_U^{\#}(g_i))$$

where $f_U^{\#} : \mathscr{O}_X(U) \to \mathscr{O}_Y(f^{-1}(U))$ is a local pullback. Thus $U_i \in \mathscr{U}$.

($\Leftarrow$) Invoke Exercise 3.

Problem (Exercise 5)**.** *Let X be a noetherian scheme and $\mathscr{F}$ a coherent sheaf on X. Prove that $\mathscr{F}$ is locally free of rank one (i.e., it is a line bundle) if and only if there exists a coherent sheaf $\mathscr{M}$ such that $\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{M} \cong \mathscr{O}_X$. Such $\mathscr{O}_X$-modules are called invertible.*

Solution.

($\Rightarrow$). This part is for the most part covered in the notes. The sheaf $\mathscr{M}$ must be a dual sheaf to $\mathscr{F}$, *i.e.*

$$\mathscr{M} = \mathscr{F}^\vee = \mathscr{H}\mathrm{om}_{\mathscr{O}_X}(\mathscr{F}, \mathscr{O}_X).$$

The isomorphism $\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{M} \to \mathscr{O}_X$ is given by $s \otimes \phi \mapsto \phi(s)$. To show it is an isomorphism, check it locally. Since $\mathscr{F}$ is a line bundle, $\mathscr{F}|_U \cong \mathscr{O}_U$ and thus $\mathscr{F}^\vee|_U \cong \mathscr{O}_U$, and thus $(\mathscr{F} \otimes \mathscr{F}^\vee)|_U \cong \mathscr{O}_U$. Clearly, $\mathscr{O}_U \otimes \mathscr{O}_U \to \mathscr{O}_U$ via $a \otimes b \mapsto ab$ is an isomorphism.

($\Leftarrow$). Now given $\mathscr{F}$ is invertible (*i.e.*, $\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{M} \cong \mathscr{O}_X$ for some coherent sheaf $\mathscr{M}$), we need to show that it is a line bundle. First, we pass to stalks. Since stalks preserve isomorphisms, we have $\mathscr{F}_x \otimes_{\mathscr{O}_{X,x}} \mathscr{M}_x \cong \mathscr{O}_{X,x}$. Here, $\mathscr{O}_{X,x}$ is the local ring (say, with maximal ideal $\mathfrak{m}$ and residue field $k = A/\mathfrak{m}$) while $F = \mathscr{F}_x$ and $M = \mathscr{M}_x$ are finitely generated A-modules.

Thus, we rewrite stalk-level isomorphism to $F \otimes_A M \cong A$. Tensor it with k to get $(F \otimes_A k) \otimes_k (M \otimes_A k) \cong k$ or $F/\mathfrak{m}F \otimes_k M/\mathfrak{m}M \cong k$. Thus, $\dim_k(F/\mathfrak{m}F) = \dim_k(M/\mathfrak{m}M) = 1$. Consequently, both F and M are generated by a single value in A. Thus $F \cong A/I$ and $M \cong A/J$ for some ideals $I, J \subseteq A$ (Nakayama Lemma). Since $F \otimes_A M \cong A$, we get $A/(I+J) \cong A$, thus $I + J = 0$ or $I = J = 0$.

Consequently, we have shown that $\mathscr{F}_x = \mathscr{O}_{X,x}$. I believe using the fact that X is noetherian, one can lift it up to an isomorphism $\mathscr{F}|_V \cong \mathscr{O}_V$.